

ATP Creativity

Isolating, Controlling, Measuring, and Optimizing Creativity in Automated Theorem Proving

Brian Tenneson

August 4, 2026

Abstract. This paper develops a controlled experimental and data-scientific framework for automated-theorem-prover creativity. Creativity is operationalized as verified structural novelty under fixed information, training, formal environment, and computational budget. The first part isolates creative search from correctness, randomness, memorization, brute force, and added computation; introduces information conditions, a creativity-control vector, proof-space scarcity, and verified creative yield. The second part turns those controls into normalized features and a scalar control index, then defines the empirical creativity score that should actually be optimized. Repeated seeded runs, held-out theorem families, response-surface modeling, and sample-efficient optimization make the problem a data-science problem. Creativity is treated as one objective among success, cost, robustness, and proof quality rather than as an unquestioned universal good.

Part I - Isolating and Controlling Creativity

1. Operational definition

Creativity is the production of a verified proof that is structurally novel under controlled information and equal resource constraints. The verifier remains deterministic: it decides correctness. Creative behavior lies in the search policy's selection and organization of legal proof actions. The motivating intuition is not that creativity contains no enumeration, but that it discovers or constructs the distinctions that make most undirected enumeration unnecessary.

2. Frozen experimental environment

Across compared runs freeze the formal system F , hypotheses Γ , target ϕ , available axioms and theorems, legal state/action spaces, verifier, model architecture and weights, training data and preprocessing, resource budgets, and certificate-normalization rules. Otherwise a supposed creativity gain may come from more knowledge, more compute, target leakage, or weaker verification.

3. Information conditions

K0: target plus frozen environment only (independent discovery). **K1:** broad theorem-family or decomposition hints (guided structural discovery). **K2:** information correlated with a known proof route (guided reconstruction). A K2 proof may be novel but should not be called an independent rediscovery without qualification.

4. Creativity controls

$$u = (\tau, w, b, n, r, \lambda, l)$$

The coordinates govern action temperature, candidate width, breadth versus depth, novelty bonus, restarts, lemma construction, and proof-length preference. Changing them may redistribute a fixed budget but may not enlarge it.

5. Creativity is not randomness

$$CI(P) = 1_{\text{verified}}(P) * \text{Independence}(P) * \text{StructuralNovelty}(P)$$

Novelty may use normalized proof-DAG distance, unfamiliar predecessor theorems, a different decomposition, reusable intermediate lemmas, different maps or constructions, bypass of expected lemmas, and diversity across seeds. Proofs must first be normalized to remove padding, detours, duplicates, and irrelevant permutations.

6. Proof counting and scarcity

Let $P_F(\Gamma, \phi, n)$ count length- n proofs, $\|\phi\|_F(\Gamma) = m$ be shortest proof length, and $P_{\min, F}(\Gamma, \phi)$ count shortest proofs. If a is maximum rule arity and κ_k counts k -ary rules, define:

$$B_F(\Gamma, n) = \text{PRODUCT}_{\{j=1..n\}} [|\Gamma| + \text{SUM}_{\{k=1..a\}} \kappa_k (j-1)^k]$$

Then $P_F(\Gamma, \phi, n) \leq B_F(\Gamma, n)$. The deliberately crude envelope gives a common combinatorial reference. Define shortest-proof scarcity:

$$R_F(\Gamma, \phi) = \log_2 [B_F(\Gamma, m) / P_{\min, F}(\Gamma, \phi)]$$

R_F is a target-relative index, not an exact entropy. Few shortest certificates imply greater scarcity relative to the same envelope.

7. Verified creative yield

For information condition K_j , training exposure p , and fixed budget T , let $D(u)$ be the probability of a verified proof and $N(u)$ the expected normalized structural novelty conditional on success.

$$Y(u; p, K_j, T) = D(u; p, K_j, T) N(u; p, K_j, T) \quad Y^*_F(u; p, K_j, T) = D(u) N(u) R_F(\Gamma, \phi)$$

The product gives little credit to fruitless noise or routine reconstruction. D and N describe the prover; R_F describes the target.

8. Isolation and repeated trials

Hold $(F, \Gamma, \phi, p, K_j, T)$ fixed and compare u with a deterministic-exploitation baseline u_0 :

$$\Delta Y^*_F(u) = Y^*_F(u) - Y^*_F(u_0)$$

Use prespecified independent seeds and record outcome, nodes, time, proof DAG, novel lemmas, and seed. Separate verified proof, honest budget exhaustion, frontier empty, and implementation fault. Estimate uncertainty across seeds and score novelty only after independent verification and normalization.

9. Expected curve and central question

A plausible curve is an inverted U: low exploration overuses familiar routes; moderate exploration supports productive branching and lemma discovery; excessive exploration becomes poorly directed and reduces success. The central question is: **At what controlled exploration level does verified structural novelty emerge most efficiently relative to the proof-space envelope?**

Part II - Measuring and Optimizing Creative Performance

10. Two scalar quantities

The controls can be summarized, but the summary is not itself measured creativity. Distinguish $\mathbf{c}_F(\mathbf{u})$, a scalar description of exploratory settings, from $\mathbf{C}_F(\mathbf{u})$, empirical creative performance on benchmark B. A maximally exploratory setting may have low creativity if it seldom proves anything.

11. Formal-system-relative scaling

Transform each raw coordinate into a behavioral feature $z_{i,F}(\mathbf{u}_i)$ in $[0,1]$. Examples are normalized action entropy, log width relative to typical legal branching, budget fraction spent on breadth or lemma construction, restart frequency, novelty influence on rankings, and tolerance for longer routes. Dependence on F is appropriate because identical numerical controls can induce different behavior in different formal systems.

12. Calibrated control index

$$\mathbf{c}_F(\mathbf{u}) = \sum_i \alpha_{i,F} z_{i,F}(\mathbf{u}_i) \quad \alpha_{i,F} \geq 0, \quad \sum_i \alpha_{i,F} = 1$$

Equal weights form a transparent baseline; ablations or predictive modeling can estimate better weights. The index is deliberately value-laden because weights encode what counts as structured exploration. It becomes statistically suspect when the same theorem outcomes select the weights and support the final claim. Publish the full feature vector, transformations, and weights; use separate calibration, validation, and test theorem sets.

13. Empirical creativity objective

$$\mathbf{C}_F, \phi(\mathbf{u}) = \mathbf{D}_F, \phi(\mathbf{u}) \cdot \mathbf{N}_F, \phi(\mathbf{u})$$